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1. Project Overview & Objective

The objective of this milestone is to develop a **deep learning classification model** capable of identifying abnormalities in **Chest X-ray images**.

The classification system predicts whether an X-ray image is:

- **Normal**
- **Abnormal**

This model acts as the **first stage in the automated medical diagnosis pipeline**, helping filter abnormal cases for further analysis.

2. Dataset Preparation

The dataset used for training contains chest X-ray images from the **VinBigData Chest X-ray dataset**.

Images were preprocessed and organized to train a deep learning classification model.

2.1 Image Preprocessing

The following preprocessing steps were applied:

- Conversion of **DICOM images to PNG format**
- Image resizing to **224 × 224 pixels**
- Normalization of pixel values
- Data augmentation for improving model generalization

These steps ensure consistent input size and improve model performance.

2.2 Dataset Organization

The dataset was organized into **training and validation folders**.

Folder Structure

classification_data/

|

├── train/

| ├── normal

| └── abnormal

|

└── val/

 ├── normal

 └── abnormal

This structure allows the **PyTorch DataLoader** to automatically label images during training.

3. Classification Model (ResNet18)

The classification model is based on **ResNet18**, a deep convolutional neural network pretrained on **ImageNet**.

Transfer learning was applied by replacing the final fully connected layer to classify chest X-ray images.

Model Architecture

Component	Specification
Base Model	ResNet18
Pretrained Weights	ImageNet
Input Size	224 × 224 RGB
Output Classes	2 (Normal, Abnormal)
Framework	PyTorch

The pretrained backbone allows the model to learn useful image features efficiently.

4. Training Configuration

The following parameters were used during training.

Parameter	Value
Loss Function	CrossEntropyLoss
Optimizer	Adam
Learning Rate	0.0003
Batch Size	32
Epochs	20
Device	GPU / CPU

The model was trained using **mini-batch gradient descent** to optimize classification accuracy.

5. Results & Performance

After training, the classification model achieved strong performance on the validation dataset.

Performance Metrics

- **Validation Accuracy:** ~95%
- **Training Loss:** Decreased consistently during training
- **Validation Loss:** Stable across epochs

The results indicate that the model can effectively distinguish between **normal and abnormal chest X-ray images**.

6. Challenges & Observations

During model development several challenges were observed:

1. Class Imbalance

The dataset contains more **normal images than abnormal images**, which may affect prediction balance.

2. Image Quality Variations

Chest X-ray images vary in brightness and contrast, requiring normalization.

3. Hardware Limitations

Training deep learning models requires **high GPU memory and computation power**.

Despite these challenges, the model successfully learned relevant patterns.

7. Conclusion

In Milestone 2, a **ResNet18-based classification model** was successfully developed and trained to detect abnormalities in chest X-ray images.

The model achieved **high validation accuracy (~95%)**, demonstrating its capability to distinguish between normal and abnormal scans.

This classification model provides the **foundation for further improvements**, including multi-disease classification and integration with detection models in future milestones.